

0.5Li₂SO₄-ZrCl₄ — experiment-structure-MLIP comparison

A sulfate-containing glass tests transport and local structure simultaneously

1 Literature

Tang et al. (2026)
Experiment + total scattering
AIMD + tuned MACE
 $\sigma_{30^\circ\text{C}} = 1.5 \text{ mS cm}^{-1}$

2 Model & software

Cluster-packed glass
Li₃₂Zr₃₂Cl₁₂₈S₁₆O₆₄
272 atoms; fixed cell
NEP89 / GPUMD

3 MD conditions

T: 320 / 330 / 340 / 350 K
Ensemble: NVT (MTTK)
 $\tau_T = 100 \text{ fs}$; $dt = 0.5 \text{ fs}$
Equil.: 50 ps
Production: 300 ps

4 Analysis

MSD $\rightarrow$ $D(T)$, σ_{NE} and E_a
RDF and coordination
Li-O environment / mobility
Experiment / tuned-MACE comparison

5 Conclusion

Sulfate motif retained
Fit can approach literature
Long-range diffusion limited